When Worst-Case Prediction Still Compresses: Tokenization and Adversarial Robustness in LLM-Based Lossless Compression

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Abstract

LLM token tables can act as global string compressors: variable-length byte sequences become fixed-width token IDs that support fast decoding and execution on encoded data. Predictive LLM compression adds a second layer, entropy-coding those IDs with next-token probabilities. We show that robustness depends on the product of both layers—tokens per source byte and coded bits per token—and cannot be inferred from either token-table compression ratio or perplexity alone. We construct canonical adversarial sequences and separately attribute tokenization gain, model surprise, arithmetic-coder loss, dictionary transfer, and framing. For Qwen2.5-0.5B, full-vocabulary adversarial sequences cost 5.73 times more bits/token than matched text8, but tokenization absorbs enough of that cost to retain a 51.3% ideal byte rate. When restricted to one-byte tokens, however, the same predictive layer expands data to a 143.4% arithmetic payload. Post-hoc vocabulary masking reduces adversarial model cost by 35.2%. These results identify tokenizer fertility as a robustness boundary and motivate co-designing token tables, predictive coders, and guarded fallback paths.

1 What is doing the compression?

"Language Modeling Is Compression" formalizes how next-symbol probabilities can drive a lossless arithmetic coder and describes tokenization itself as lossless pre-compression (Delétang et al., 2024). This observation implies that so-called LLM compression combines two distinct mechanisms: a tokenizer replaces variable-length source strings with token IDs, and a language model predicts those IDs so that an entropy coder can shorten likely ones. LLMZip is another instance of this composed design (Valmeekam et al., 2023).

Schmidt et al. (Schmidt et al., 2026) isolate the first mechanism. They use an LLM vocabulary as a global symbol table in which common byte strings map to fixed-width IDs and one-byte entries guarantee coverage. Unlike a block-local code, this stable representation supports equality checks, hashing, joins, and aggregations before decoding. These properties make token tables particularly attractive for databases, where strings can remain encoded during common relational operations and be decoded only when needed. Thus tokenization can be a useful compression primitive without language-model inference.

What remains unclear is how much of an LLM codec's behavior comes from this token representation and how much comes from prediction. Delétang et al. compare ASCII and BPE on natural enwik9, but the two effects have not been isolated under inputs chosen against realized end-to-end size. This matters because prediction loss is measured per token while compression is measured per source byte: a poor probability model may still sit inside a pipeline that compresses when its tokens represent many bytes, whereas low-fertility tokenization can expose expansion. We therefore study the tokenizer–predictor–coder pipeline as a two-layer system.

Contributions. We (i) factor end-to-end rate into tokenizer fertility and predictive coding cost; (ii) construct lossless attacks against token-, byte-, and realized-code objectives; and (iii) use matched natural and adversarial inputs to show when the token layer absorbs predictive surprise and when it exposes expansion.

Scope. We hold one pretrained tokenizer/model fixed and vary canonical inputs adversarially. Our evidence is a Qwen2.5-0.5B case study and does not contradict the token-table paper's corpus-level results.

2 A two-layer compression model

Let s be UTF-8 text, $T(s) = x_{1:N}$ its tokens, $\mathrm{decode}(x)$ its decoded byte string, $h_t = x_{1:t-1}$, and $p_\theta(v \mid h_t)$ the model distribution. If the codec retains dictionary $A \subseteq V$, it renormalizes that distribution to $q_\theta(v \mid h_t, A)$. The realized sequence NLL is

$$L_A(x) = -\sum_{t=2}^{N} \log_2 q_\theta(x_t \mid h_t, A). \tag{1}$$

This is sequence cross-entropy, not Shannon entropy of the predictive distribution. By the source-coding correspondence, an ideal arithmetic code has length approximately equal to this NLL (Shannon, 1948; Witten et al., 1987; Delétang et al., 2024).

2.1 Factorizing the two layers

Consider a sequence of N tokens that decodes to U source bytes. Its *fertility*, $f(x) = U/N$, is the average number of bytes represented by one token. A fixed-width token table spends $w = \lceil \log_2 |V| \rceil$ bits per token, or Nw bits in total. Relative to the $8U$ bits in the original bytes, its rate is $R_T = Nw/(8U) = w/(8f)$. The predictive coder replaces the fixed cost w with an average cost $\bar{\ell}_A = L_A/(N-1)$ bits per predicted token. Its ideal rate is therefore

$$R_{\mathrm{pred}} = \frac{L_A}{8U} \approx \frac{\bar{\ell}_A}{8f(x)} = \underbrace{\frac{N}{U}}_{\text{tokens/source byte}} \underbrace{\frac{\bar{\ell}_A}{8}}_{\text{coded bytes/token}}. \tag{2}$$

The predictor improves the token table when $\bar{\ell}_A < w$ and makes its token stream larger when $\bar{\ell}_A > w$. Yet the end-to-end output can still compress if fertility f is large enough. This separates the token-table paper question —how many fixed-width IDs represent the bytes?—from ours—how costly are those IDs under the model?

For arithmetic payload C_{AC}, dictionary cost C_A, and header cost C_H, the realized file rate is

$$R_{\mathrm{file}} = \frac{C_{\mathrm{AC}} + C_A + C_H}{8U}. \tag{3}$$

The gaps from Eq. 2 attribute frequency quantization/coder, dictionary, and framing overhead. Values above one denote expansion. We report the raw offline rate and treat model weights as shared decoder state; an adjusted two-part code would additionally charge their size (Delétang et al., 2024).

Compression-oriented objectives. Worst-case prediction loss is not worst-case compression: tokens represent unequal amounts of source data. For actual serialized length C_{actual} in bits, the end-to-end adversary is

$$\max_x \frac{C_{\mathrm{actual}}(x)}{8|\mathrm{decode}(x)|_{\mathrm{bytes}}}. \tag{4}$$

Even a terrible probability model can therefore belong to a pipeline that still compresses because variable-length tokenization is favorable. A cheap greedy approximation chooses high-surprisal tokens representing few decoded bytes:

$$x_t = \arg\max_{v \in V} \frac{-\log_2 p_\theta(v \mid x_{<t})}{|\mathrm{decode}(v)|_{\mathrm{bytes}}}. \tag{5}$$

Because standalone decoding can be context-dependent, our implementation uses $\Delta_b(v;h) = |\mathrm{decode}(h \parallel v)|_{\mathrm{UTF8}} - |\mathrm{decode}(h)|_{\mathrm{UTF8}}$ in the denominator. Thus minimum probability and maximum surprisal per decoded byte can rank candidates differently.

3 Threat model and attacks

Adversary. Use a white-box adversary with access to tokenizer, logits, and coding algorithm. The adversary chooses a length-N token sequence after a fixed seed, subject to a candidate alphabet G. Because arbitrary token IDs can change after decoding and retokenizing, we accept a prefix only when $T(\mathrm{decode}(x_{1:t})) = x_{1:t}$; final decoding must recover the same IDs and bytes exactly. We evaluate the full non-special vocabulary, printable ASCII, and canonical one-byte ASCII. The last isolates predictive failure from token-length variation; broader alphabets expose the tokenizer interface.

Attack variants.

Variant	How the next token or sequence is chosen
Random token	Seeded uniform sampling among valid extensions; no model-based objective.
Minimum probability	Greedily choose the valid token with smallest $p_\theta(v \mid x_{<t})$.
Surprisal/byte	Greedily maximize $-\log_2 p_\theta(v \mid x_{<t})/\Delta_b(v; x_{<t})$ (Eq. 5).
Serialized-size search	Beam search that clones coder state and maximizes realized bits per decoded byte (Eq. 4).

Controls are ordinary text8, seeded random valid UTF-8, and unrestricted random bytes for byte-capable codecs. We apply gzip, Zstandard, and Brotli to every exact decoded sequence. Beam search is approximate unless its branch factor covers all candidates; changing beam ancestry disables the KV cache.

Global mask. We construct each adversarial sequence once under the full distribution, then replay identical IDs and histories under its occurring-token mask. This isolates renormalization and is not a mask-adaptive attack.

4 Experimental evaluation

Setup. We evaluate Qwen2.5-0.5B with its 151,643-token vocabulary. The adversary creates three 1,000-token sequences from starting token IDs 785, 32, and 641 by repeatedly choosing the lowest-probability canonical extension. Every completed sequence satisfies $T(\mathrm{decode}(x)) = x$ and decodes to 6,988, 6,812, and 6,943 UTF-8 bytes. We then replay the identical token IDs and contexts using the set of tokens occurring in each completed sequence; the three post-hoc dictionaries contain 592, 606, and 587 tokens. Thus masking changes only probability normalization, not generation.

The natural-text control is the first 100,000 text8 tokens, processed by the same model with context length 1,000, retained context 100, KV caching, and arithmetic coding. It uses 16 independently seeded batches, so 99,984 tokens are predicted. We report mean adversarial surprisal with sample standard deviation across the three starts. Text8 is one deterministic matched run per dictionary policy and therefore has no error bar.

Token-level failure, byte-level buffering. Figure 1 shows that full-vocabulary adversarial inputs cost 28.409 ± 0.077 ideal bits per predicted token, $5.734\times$ text8 at 4.954. Because Qwen has 151,643 symbols, a fixed ID needs 18 bits; the predictive layer therefore makes this adversarial token stream 58% larger than fixed-width IDs. However, the sequences decode to an average 6.914 source bytes/token. This strong token-table compression buffers model surprise, leaving an ideal end-to-end rate of 51.346% of raw UTF-8 rather than expansion.

Post-hoc occurring-token dictionaries contain 592, 606, and 587 symbols. Masking the same IDs and histories lowers adversarial cost to 18.402 ± 0.833 bits/token, a 35.2% reduction, and the byte-normalized floor to 33.284%. It remains $3.806\times$ the matched masked-text8 cost of 4.835 bits/token. The identical full-model scores under both policies verify that only normalization changes.

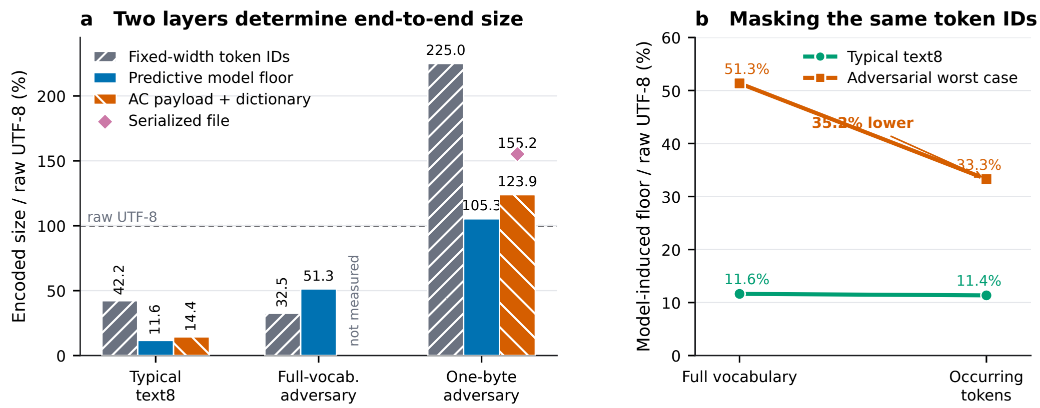

Figure 1: How tokenization and predictive coding jointly determine size for Qwen2.5-0.5B. (a) Fixed-width token-table rate, model-induced floor, and realized coding costs relative to each input's UTF-8 bytes. The full-vocabulary adversarial arithmetic payload has not been measured. (b) Post-hoc masking rescoring of identical token IDs and histories; masking reduces the adversarial floor by 35.2%.

Typical text and realized overhead. For text8, model-induced floors are 11.631% of raw UTF-8 with the full vocabulary and 11.352% with occurring tokens. Finite coding and metadata raise realized totals to 14.405% and 13.434%, or 6.942× and 7.444× compression. The occurring-token bitmap costs 10,818 bytes (2.03% of raw size), less than its coding gain at 100,000 tokens.

Removing tokenization gain exposes expansion. The notebook one-byte-ASCII test fixes fertility at one byte/token. Greedy minimum probability and surprisal-per-byte then coincide at a 135.48% model floor and 143.36% arithmetic payload; an NLL beam reaches 136.70% and 144.14%. Thus predictive coding improves on 18-bit fixed Qwen IDs but cannot beat the original 8-bit bytes. The narrow 32-token beam validates this mechanism and the accounting, not global optimality. Together, the experiments show both regimes of Eq. 2: multi-byte tokens can absorb extreme surprise, while low-fertility inputs expose predictive expansion.

5 Systems implications and limitations

The systems lesson is to preserve token IDs as the global, queryable in-memory representation and treat predictive arithmetic coding as an optional block-level storage or transport layer. Source-normalized accounting identifies whether fragility comes from the tokenizer, predictor, quantizer, dictionary, or container. For untrusted blocks, a guard can store the smaller of predictive and conventional encodings plus a selector bit, bounding size by the fallback plus one bit; its decoding granularity, latency, and throughput remain to be measured.

Our evidence is limited to one model, synthetic white-box inputs, three starts, finite context, approximate search, and one coder/container. It does not establish global worst cases or generalize to multilingual text or code. The attack could enable storage or compute amplification; practical mitigations include size caps, early-abort expansion detection, bounded inference, and fallback codecs.

6 Conclusion

Robust learned compression is an end-to-end systems problem. Canonical byte-aware attacks and explicit cost attribution show that the tested adversarial sequences are 3.81–5.73 times harder than matched text8, while guarded fallback offers a principled path from diagnosis to robust operation.

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